EDS 223 Homework #1

Geospatial Analysis & Remote Sensing

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Table of contents

Loading in packages: 1
Data Wrangling and Filtering 1
Map #1: Orange County PM 2.5 Levels (Annual Average) 2
Map #2: Percentage of People of Color in Orange County 3
Map Interpretation: 4
Data Citations: 5

Loading in packages:

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(spData)
library(viridisLite)
```

Data Wrangling and Filtering

```
# Read in the database of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNI
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/swaggerboy/Documents/EDS223-hw/EDS223-HW1/data/ejscreen/EJSCREEN_2023_BG_StatePct_w
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```r
# Filter to California
california <- ejscreen %>%
    dplyr::filter(ST_ABBREV == "CA")

# Filter to Orange County
orange_county <- ejscreen %>%
    dplyr::filter(CNTY_NAME %in% c("Orange County"))

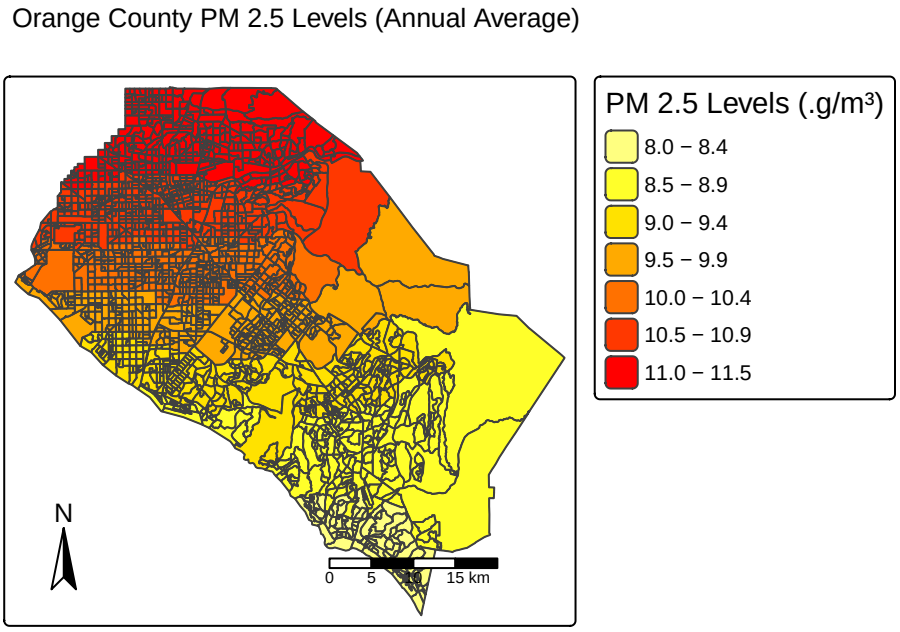
# Find the average values for all variables within counties
california_counties <- aggregate(california, by = list(california$CNTY_NAME), FUN = mean)

# Narrow down to Orange County
oc <- california |>
    filter(
        CNTY_NAME == 'Orange County'
    )
# Remove missing data row (ocean)
oc_clean <- oc[-(2049),]
```

Map #1: Orange County PM 2.5 Levels (Annual Average)

```r
# Basemap: Orange County boundary
tm_shape(oc_clean) +
    # Fill in city polygons with 'PM25' data
    tm_polygons(
        fill = 'PM25',
        fill.scale = tm_scale(values = rev(heat.colors(5))),
        fill.legend = tm_legend(title = 'PM 2.5 Levels ( g/m³)'
                                )) +
    # Add scalebar
    tm_scalebar() +
```

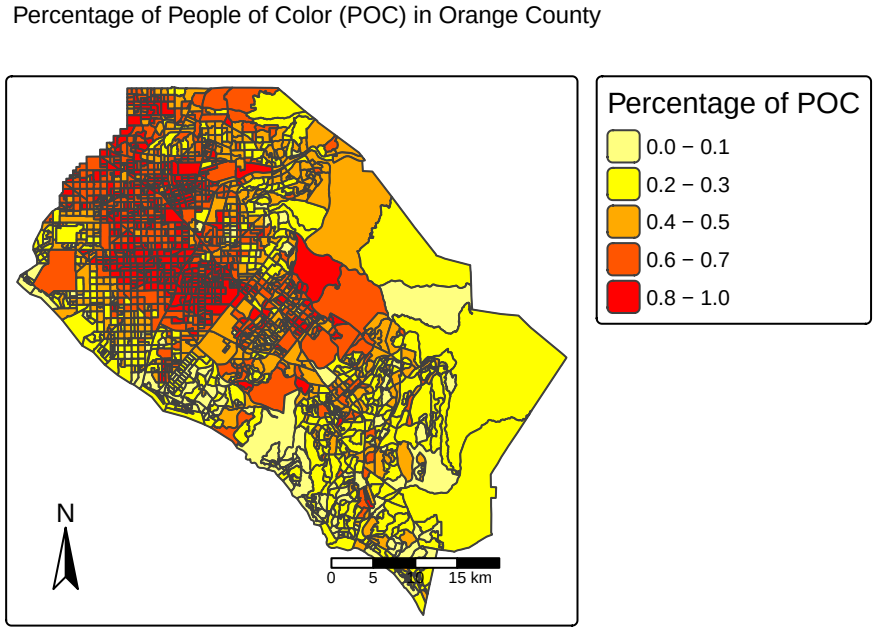
```
    # Add compass with only North arrow
    tm_compass(position = c('bottom', 'left')) +
    # Adjust border line width
    tm_borders(lwd = 0.05) +
    # Add title
    tm_title("Orange County PM 2.5 Levels (Annual Average)")
```



Map #2: Percentage of People of Color in Orange County

```
# Basemap: Orange County boundary
tm_shape(oc_clean) +
    # Fill in city polygons with 'PEOPCOLORPCT' data
    tm_polygons(
        fill = 'PEOPCOLORPCT',
        fill.scale = tm_scale(values = rev(heat.colors(5))),
        fill.legend = tm_legend(title = 'Percentage of POC')) +
    # Add scalebar
    tm_scalebar() +
    # Add compass
    tm_compass(position = c('bottom', 'left')) +
```

```
    # Adjust border line width
    tm_borders(lwd = 0.05) +
    # Add title
    tm_title("Percentage of People of Color (POC) in Orange County")
```



Map Interpretation:

- **PM25**: Levels of PM 2.5 measured using an annual average.
- **PEOPCOLORPCT**: Percent of people in a census block who designate their racial status as a race other than white alone or list their ethnicity as hispanic/latino.

Map #1 displays high annual levels of PM 2.5 concentrated primarily within the Northern most census blocks while Southern census blocks experience lower levels. Similarly, Map #2 displays high percentages of POC in the Northern to middle census blocks and lower percentages in the South. The data within these maps suggest that areas within Orange County with high percentages of POC communites are exposed to high levels of PM 2.5 more than areas with low percentages of POC communities.

Data Citations:

U.S. Environmental Protection Agency (EPA), 2023. EJScreen Technical Documentation. Accessed: October 8th, 2025. **Link:** https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcI